Baihan Zhang

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Education

Stanford University Stanford, CA
M.S. & B.S. in Computer Science *Exp. Graduation Spring 2027*

- GPA: 3.7/4.0
- **Relevant Coursework:** Machine Learning, Collaborative Robotics, Cryptography, Quantum Computing, Post-Quantum Cryptography, Visual Computing Systems, Linear Algebra.
- **Activities:** Columnist at Stanford Blockchain Review, Member of Stanford AI Alignment, Member of Stanford Quantum Computing Association

Experience

Stanford Open Data Portal Stanford, CA
Full-Stack Engineer & Platform Lead *Jan 2026 – Present*

- Spearheading the end-to-end development of "Doppelganger," a conversational AI platform that replaces static surveys with dynamic "digital twins" that evolve through natural voice interaction.
- Architected a low-latency voice processing pipeline (Record → Transcribe → LLM Inference) using Node.js and Vite, optimizing the feedback loop to maintain conversation immersion.
- Engineered a novel memory conditioning system with progressive personality weighting to prevent LLM hallucinations, ensuring consistent identity modeling across long-term user interactions.
- Designed a JWT-secured backend architecture with persistent conversation storage, enabling multi-agent simulations where AI twins interact independently in a virtual "playground."

Stanford University (Kundaje Lab) Stanford, CA
Research Assistant (ML Systems & Interpretability) *Jan 2025 – Dec 2025*

- Engineered interpretable deep learning pipelines using TensorFlow and PyTorch; utilized tools like TF-MODISCo and DeepSHAP to analyze model decision boundaries on high-dimensional genomic data.
- Reverse-engineered deep learning decision processes to assign feature contribution scores, validating the model's biological causal reasoning.
- Built and trained custom CNN architectures (chromBPNet), optimizing model performance for large-scale biological sequence processing (chromatin accessibility prediction).
- Identified two specific proteins with high attribution scores correlating to thyroid cancer occurrence, optimizing model accuracy for scientific discovery.

Herbal Decode (Princeton Non-Profit) Princeton, NJ
Machine Learning Engineer Intern *Aug 2025 – Oct 2025*

- Built a production RAG (Retrieval-Augmented Generation) pipeline using BGE-M3 embeddings and Qdrant (HNSW indexing) to process complex Traditional Chinese Medicine texts.
- Implemented an efficient hybrid data ingestion layer using direct text extraction with PaddleOCR fallbacks, handling unstructured data at scale.
- Achieved sub-second semantic search response times across 65,000+ document chunks by optimizing HNSW indexing parameters to balance recall and latency.
- Designed end-to-end ETL pipelines to standardize data formats for vector storage and assessed feasibility of OCR tools for low-quality source documents.

FANUC (Industrial Robotics Company) Detroit, MI
Product Management Intern *March 2025 – June 2025*

- Spearheaded 90+ customer interviews and market research that reshaped corporate strategy, secured approval, and launched a seed financing program providing FANUC co-bots to pre-seed/Series A robotics startups.
- Assisted senior leaders with the assessment of project costs and usage scenarios, conducting feasibility analysis to validate business use cases.
- Served as a liaison between engineering and business teams, ensuring technical specifications aligned with customer scenarios.

Projects

Sparse Neural Network Architecture Research | *Python, PyTorch, COMET* Nov 2025

- Developed a diagnostic framework for Conditionally-Overlapping Mixture-of-Experts (COMET) models to analyze routing stability, routing collapse, and computational efficiency.
- Formulated novel ”Mask Coherence” and ”Feature Sensitivity” metrics to quantify semantic fragmentation and systematic robustness failure modes.
- Demonstrated via mathematical analysis that background noise, rather than semantic content, drove routing decisions in CIFAR-10 classifiers, exposing critical robustness failures in sparse architectures.

Website Optimizer for On-Chain Agents (YC Hackathon Top 10) | *React, AI Agents* Nov 2025

- Built agentic infrastructure converting e-commerce sites into structured formats (commerce.txt) for autonomous agent access.
- Designed a standardized data format to structure e-commerce information, establishing a clear schema for machine interaction and payment execution.
- Placed Top 10 out of 150 teams, selected to pitch directly to Y Combinator and Stripe partners.

Quantum Circuit Analysis (BlueQubit Hackathon Winner) | *Python, Qiskit, OpenQASM* Nov 2025

- Secured prize ($150 BTC) by solving the ”One Peak” challenge: identifying max-amplitude bitstrings in increasingly complex OpenQASM 2.0 circuits.
- Engineered a Qiskit analysis script to parse qasm2 files and perform state vector simulations, effectively isolating the peak probability outcome.
- Optimized the search space for ”peak bitstrings” by analyzing circuit amplitude distributions, effectively reverse-engineering the circuit logic.

Ethereum Splitwise & Multisig | *Solidity, Foundry, Graph Theory* Oct 2025

- Developed gas-optimized smart contracts implementing on-chain debt graphs with automatic cycle resolution algorithms.
- Verified system correctness and security behaviors through extensive Foundry unit tests, ensuring the immutability and safety of fund management logic.

FilmR: CLIP-Based Video Retrieval System | *PyTorch, CLIP, Computer Vision* March 2025 – June 2025

- Benchmarked a video retrieval system against baseline CLIP models under severe compute and data constraints.
- Achieved a 2.6% average gain in embedding diversity while maintaining object detection accuracy across 40+ complex queries.

Plant Health Detector | *TensorFlow, CNNs, Computer Vision* Jan 2025 – March 2025

- Designed a custom weighted CNN to detect diseased strawberry leaves, achieving 98.8% accuracy and outperforming Kaggle baselines by 4%.
- Curated a novel dataset by collecting and hand-labeling 800+ real-world farm images to improve model generalization.

Technical Skills

Languages: C, C++, Python, TypeScript, JavaScript, Solidity, SQL, HTML/CSS
ML/AI: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, LangChain, RAG, Qiskit, Embeddings, Transformers
Systems & Tools: Linux, Docker, Git, GDB, Bash, MongoDB, PostgreSQL, Qdrant, Foundry, AWS, LaTeX, Figma
Data Engineering: ETL Pipelines, Schema Design, Data Lineage, Data Cleaning, JSON/BSON, Data Anonymization
Concepts: Distributed Systems, Low-Level Programming, Memory Safety, Cryptography, Quantum Algorithms, Graph Theory